Summary

Microorganisms need macro and micronutrients for their growth. Based on the energy source, organisms are grouped into Phototrophs and Chemotrophs. Based on carbon source, they are classified into autotrophs and heterotrophs. Organisms are grouped into lithotrophs and organotrophs based on their electron source. The four nutritional classes of microbes are photoautotrophs, Photoheterotrophs, chemoautotrophs and chemoheterotrophs.

Cyanobacteria are prokaryotes that can perform photosynthesis. Chlorophyll is the pigment needed to capture light energy (photons). In cyanobacteria and green plants, non cyclic photophosphorylation takes place to generate ATP and NADPH during photosynthesis whereas cyclic photophosphorylation takes place in purple and green bacteria involving only one Photosystem (PS I).

In a batch culture, bacteria show a characteristic growth pattern which consists of lag phase, log phase and stationary phase and decline phase. In a chemostat, cultures can be maintained in an exponential phase for long periods. The most important factors affecting microbial growth are temperature, pH and oxygen level. Total count and viable count are the two widely used methods to measure cell numbers.

Evaluation

Multiple choice questions

1. An example of photoautotroph
   a. Cyanobacteria
   b. Algae
   c. Green plants
   d. All of the above

EPTP1E

2. Magnesium is needed
   a. For cell wall synthesis
   b. As cofactor for enzymes
   c. For photosynthesis
   d. For protein synthesis
3. One of the following is an example for chemoautotroph
   a. Cyanobacteria
   b. Purple and green non sulphur bacteria
   c. Iron bacteria
   d. Protozoa
4. The phase of growth in which the growth rate is equal to the death rate is
   a. Stationary phase
   b. Death phase
   c. Exponential phase
   d. Lag phase
5. Organisms that are capable of growing in 0°C are called
   a. Thermophiles
   b. Hyper thermophiles
   c. Barophiles
   d. Psychrophiles
6. Halophiles are organisms that can grow in
   a. Low water activity
   b. High salt concentration
   c. Low temperature
   d. High pH
7. An example of microaerophilic organism is

   a. Bacillus *b. Azospirillum*
   c. Pseudomonas *d. Escherichia.coli*

TN_GOVT_XI_Micro_Biology_CH06.indd 84 16-02-2019 9.58.15 AM